CORE PURE / VECTORS, MATRICES & LINEAR TRANSFORMATIONS

Plane Intersections

Mark scheme 7 questions · 66 marks

Questions

Question	Marks by part	Marks	Page
Question 1	(a) 2 (b) 3	5	4
Question 2	(a) 2 (b) 3 (c) 3	8	5
Question 3	(a) 6 (b) 2	8	6
Question 4	(a) 2 (b) 5 (c)(i) 2 (c)(ii) 2	11	7
Question 5	(a) 4 (b) 6	10	9
Question 6	(a)(i) 2 (a)(ii) 3 (b) 5	10	10
Question 7	(a) 2 (b)(i) 6 (b)(ii) 3 (c)(i) 1 (c)(ii) 2	14	11
	Total	**66**	

Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes.

ft	follow through	**AG**	answer given in the question
cao	correct answer only, with no follow-through	**cso**	correct solution only: no errors in the part
oe	or an equivalent form	**awrt**	any answer that rounds to this value
soi	seen, or implied by later working	**isw**	ignore working after a correct answer
SC	special case: credit outside the main method	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- **[*n*]** ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

Value of g

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

Question	Scheme	Marks	Notes
1(a)	$\det \mathbf{A} = 3(-7-0) + 2(-14+28) + 1(0-7)$	**M1**	Expands along any row or column, with three 2×2 minors and their signs. Condone one slip in a minor.
	$= -21 + 28 - 7 = 0$ so $\mathbf{A}$ is singular and there is no unique solution	**A1 AG**	Correct expansion leading without error to 0, then both conclusions, singular and no unique solution.
		[2]	
1(b)	$(3x - 2y + z) + 2(2x + y - 4z) = 5 + 2(1)$	**M1**	Combines two equations to compare with the third, or solves two and substitutes into the third. Condone arithmetic slips.
	$\Rightarrow 7x - 7z = 7$, the third equation, so the system is consistent	**A1**	Correct combination matching the third equation, constants included, and consistent stated.
	The three planes meet in a common line (a sheaf) $\left[\mathbf{r} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}\right]$	**E1**	States the planes meet in a common line, or form a sheaf. Not for infinitely many solutions alone; the line's equation is not needed.
		[3]	
	Total for Question 1	**5**	

Question	Scheme	Marks	Notes
2(a)	$\det \mathbf{A} = 1(-7) - 2(-6) - 1(5) = 0$	**B1**	$\det \mathbf{A} = 0$ for the coefficient matrix; a calculator value is accepted.
	A is singular, so no unique solution	**E1**	States that **A** is singular or not invertible, so there is no unique solution. Needs $\det \mathbf{A} = 0$.
		[2]	
2(b)	$(x + 2y - z) + (2x - y + 4z) = 3 + 1$	**M1**	Adds the first two equations, or eliminates one variable from the system by any method, and compares with the third.
	$\Rightarrow 3x + y + 3z = 4$, the third equation, so consistent	**A1**	Correct sum identified with the third equation, or $0 = 0$ by elimination, and a conclusion of consistency.
	The three planes meet in a common line (a sheaf)	**E1**	States the planes meet in a single common line, or form a sheaf. Not for infinitely many solutions alone.
		[3]	
2(c)	First two equations $\Rightarrow 3x + y + 3z = 4$	**M1**	Combines the first two equations to give $3x + y + 3z = 4$, or eliminates by any method. May be seen in (b).
	$3x + y + 3z = a \neq 4$: contradiction, so inconsistent	**A1**	Contradiction with the third equation for $a \neq 4$ and a conclusion of inconsistency.
	Normals $(1, 2, -1)$, $(2, -1, 4)$, $(3, 1, 3)$: no two parallel $\Rightarrow$ triangular prism	**E1**	States triangular prism, with a reason no two planes are parallel, such as no two normals being multiples.
		[3]	
	Total for Question 2	**8**	

Question	Scheme	Marks	Notes
3(a)	$\begin{vmatrix} 1 & -1 & 2 \\ 2 & 1 & 1 \\ 3 & 0 & m+1 \end{vmatrix}$ $= 1(m+1) + 1\big(2(m+1) - 3\big) + 2(0-3)$	**M1**	Expands the determinant of the coefficient matrix in terms of m by any row or column. Condone sign slips.
	$= 3(m-2)$	**A1**	$3(m-2)$ oe, such as $3m-6$.
	$m \neq 2 \Rightarrow \det \neq 0$: the planes meet in a single point	**E1**	Relates the non-zero determinant to a unique point of intersection. The point need not be found.
	$m = 2$: first + second gives $3x + 3z = 7$ third: $3x + 3z = 7$	**M1**	Substitutes $m = 2$ and tests consistency, by combining the first two equations to compare with the third, or by elimination.
	Third equation is the sum of the first two, so consistent	**A1**	Third equation shown to be the sum of the first two, or elimination reaching $0 = 0$, with consistency stated or implied.
	The planes meet in a common line (a sheaf)	**E1**	States the planes share one common line, or form a sheaf, from a correct consistency result. The line need not be found.
		[6]	
3(b)	$\begin{vmatrix} 1 & -1 & 2 \\ 2 & 1 & 1 \\ 3 & 0 & 4 \end{vmatrix} = 1(4) + 1(8-3) + 2(0-3)$	**M1**	Forms the determinant or scalar triple product of the three vectors and attempts to evaluate it, or uses another valid coplanarity test. Condone sign slips.
	$= 3 \neq 0$, so not coplanar: the student is incorrect	**A1**	Value ± 3, depending on order, and concludes the student is incorrect.
		[2]	
	Total for Question 3	**8**	

Question	Scheme	Marks	Notes
4(a)	$\det \mathbf{M} = 1(k+1) - 1(3-2) + 0$	**M1**	Expands the 3×3 determinant along any row or column, with every nonzero 2×2 minor attempted. Condone sign slips.
	$\det \mathbf{M} = k$, so an inverse exists for $k \neq 0$	**A1**	
		[2]	
4(b)	$\mathbf{M}^{-1} = \frac{1}{2}\begin{pmatrix} 3 & -1 & 1 \\ -1 & 1 & -1 \\ -7 & 3 & -1 \end{pmatrix}$	**B1**	Correct inverse with $k = 2$, from a calculator or otherwise.
	$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \text{"}\mathbf{M}^{-1}\text{"} \begin{pmatrix} 1 \\ p \\ 4 \end{pmatrix} = \ldots$	**M1**	Multiplies their 3×3 inverse by $\begin{pmatrix} 1 \\ p \\ 4 \end{pmatrix}$, in that order, for at least one component.
	$x = \frac{7-p}{2}$	**A1**	oe. Coordinates may be given as a column vector.
	$y = \frac{p-5}{2}$	**A1**	
	$z = \frac{3p-11}{2}$	**A1**	
	Alternative: Elimination		
	$y = 1 - x \Rightarrow 3x + 2(1-x) + z = p$ $2x - (1-x) + z = 4$	**M1**	Eliminates one variable from two pairs of the equations, giving two equations in the other two. Condone sign slips.
	$x + z = p - 2, \quad 3x + z = 5$	**A1**	oe, or the correct pair for another eliminated variable.
	$2x = 5 - (p-2) \Rightarrow x = \ldots$ $y = 1 - x = \ldots, \quad z = 5 - 3x = \ldots$	**dM1**	Solves their pair for one variable and substitutes back to find the other two.
	$x = \frac{7-p}{2}$	**A1**	oe. Coordinates may be given as a column vector.
	$y = \frac{p-5}{2}, \quad z = \frac{3p-11}{2}$	**A1**	Both correct, oe.
		[5]	
4(c)(i)	$(3x + z) - (x + y) = 2x - y + z \Rightarrow q - 1 = 4$	**M1**	Combines two of the equations to match the variable terms of the third, or eliminates variables to compare with it, reaching an equation in q. Condone sign slips.
	$q = 5$	**A1**	
		[2]	

Question 4 continued

Question	Scheme	Marks	Notes
4(c)(ii)	Third equation follows from the first two, so infinitely many solutions	**E1**	States that one equation is a combination of the other two, or that the system is consistent with no unique solution, and so has infinitely many solutions.
	The three planes meet in a common line (a sheaf)	**E1**	States that the planes meet in a single common line; "sheaf" alone accepted. The line's equation is not needed.
		[2]	
	Total for Question 4	**11**	

Question	Scheme	Marks	Notes
5(a)	$\begin{vmatrix} 1 & -1 & 2 \\ k+1 & k & 2-k \\ 5-2k & 2-2k & k+2 \end{vmatrix} = 0$ $\Rightarrow k(k+2) - (2-k)(2-2k) + \ldots = 0$	**M1**	Sets the determinant of the coefficient matrix to zero and expands it, with at least two of the three 2×2 terms correct in form. Condone sign slips.
	$-2k^2 + 10k - 8 = 0$	**A1**	Correct quadratic, oe such as $k^2 - 5k + 4 = 0$.
	$-2(k-1)(k-4) = 0 \Rightarrow k = \ldots$	**dM1**	Solves their three-term quadratic by any means, a calculator allowed.
	$k = 1$ or $k = 4$	**A1**	Both values and no others.
		[4]	
5(b)	$k = 1$: $(x - y + 2z) + (2x + y + z) = 4 + 3$ $\Rightarrow 3x + 3z = 7$	**M1**	Substitutes $k = 1$ and combines two equations to compare with the third, or eliminates a variable from two pairs and compares.
	Same as the third equation, so consistent	**A1**	Shows the third equation is the sum of the first two and states consistent.
	$\mathbf{n}_1 = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$, $\mathbf{n}_2 = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$ not parallel $\Rightarrow$ the planes meet in a common line: a sheaf	**E1**	States sheaf, or a common line, with at least two normals shown not parallel.
	$k = 4$: $2(x - y + 2z) - (5x + 4y - 2z) = 2 \times 4 - 0$ $\Rightarrow -3x - 6y + 6z = 8$	**M1**	Substitutes $k = 4$ and combines two equations to compare with the third, or eliminates a variable from two pairs and compares. Condone arithmetic slips.
	But $-3x - 6y + 6z = 4$: $8 \neq 4$, so inconsistent	**A1**	Correct contradiction and states inconsistent.
	$\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}, \begin{pmatrix} 5 \\ 4 \\ -2 \end{pmatrix}, \begin{pmatrix} -3 \\ -6 \\ 6 \end{pmatrix}$: no two parallel $\Rightarrow$ triangular prism	**E1**	States triangular prism, with all three pairs of normals shown or stated not parallel. E0 without the normals.
		[6]	
	Total for Question 5	**10**	

Question	Scheme	Marks	Notes
6(a)(i)	$\det \mathbf{A} = 1(2-b) - a(4-3) = 0$	**M1**	Sets an expansion of the determinant of the coefficient matrix equal to zero. Condone one sign slip.
	$b = 2 - a$	**A1**	oe
		[2]	
6(a)(ii)	$x = -1 - ay, \quad z = 5 + (2a-1)y$	**M1**	Finds the line of intersection of two planes, or a point on it, or reduces to two equations in two variables, or writes one plane as a combination of the others. Condone sign slips.
	$3(-1-ay) + (2-a)y + 2\big(5 + (2a-1)y\big) = c$	**dM1**	Uses their b in terms of a with their line, point, reduced equations or combination to find c.
	$c = 7$	**A1**	
		[3]	
6(b)	$x = -1 - ay \Rightarrow 2(-1-ay) + y + z = 3$	**M1**	Eliminates a variable between two of the equations, giving one variable in terms of the others. Condone sign slips.
	$z = 5 + (2a-1)y$	**A1**	oe, or the correct result of another elimination.
	$3(-1-ay) + (1-a)y + 2\big(5 + (2a-1)y\big) = 5$ $\Rightarrow 7 - y = 5 \Rightarrow y = \ldots$	**dM1**	Substitutes their expressions into the third equation, with $b = 1 - a$ and $c = 5$, and solves for the remaining variable.
	$y = 2$	**A1**	
	$x = -1 - 2a, \quad z = 4a + 3$	**A1**	Both, oe, as coordinates or separate values.
		[5]	
Total for Question 6		**10**	

Question	Scheme	Marks	Notes
7(a)	$\det \mathbf{A} = 3(3-0) - 2(1-0) + a(-1-6)$	**M1**	Expands along any row or column, three terms each an element times its 2×2 determinant, or uses Sarrus' rule. Condone sign slips.
	$\det \mathbf{A} = 7(1-a)$	**A1**	oe, such as $7 - 7a$.
		[2]	
7(b)(i)	Minors: $\begin{pmatrix} 3 & 1 & -7 \\ 2+a & 3-2a & -7 \\ -3a & -a & 7 \end{pmatrix}$	**M1**	Attempts all nine 2×2 minors; implied by a matrix of cofactors.
	$\operatorname{adj} \mathbf{A} = \begin{pmatrix} 3 & -a-2 & -3a \\ -1 & 3-2a & a \\ -7 & 7 & 7 \end{pmatrix}$	**M1**	Applies the alternating signs and transposes their matrix of minors, in either order.
	$\mathbf{A}^{-1} = \frac{1}{7(1-a)} \begin{pmatrix} 3 & -a-2 & -3a \\ -1 & 3-2a & a \\ -7 & 7 & 7 \end{pmatrix}$	**M1 A1**	**M1** Divides their adjugate by their determinant from (a), in terms of a. **A1** Correct inverse, oe; the factor may be multiplied in.
	$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \text{"}\mathbf{A}^{-1}\text{"} \begin{pmatrix} 8 \\ 5 \\ 3 \end{pmatrix}$ $= \frac{1}{7(1-a)} \begin{pmatrix} 14-14a \\ 7-7a \\ 0 \end{pmatrix}$	**M1**	Multiplies their $\mathbf{A}^{-1}$, in terms of a, by $(8,5,3)^T$ in that order, at least one element attempted.
	$x = 2,\ y = 1,\ z = 0$	**A1**	cso; all three values.
		[6]	
7(b)(ii)	With $a = 1$, adds the second and third equations: $(x + 3y) + (2x - y + z) = 5 + 3$	**M1**	Sets $a = 1$ and compares a combination of two equations with the third.
	$\Rightarrow 3x + 2y + z = 8$, the first equation, so consistent	**A1**	Shows the first equation is the sum of the others, oe, and concludes the equations are consistent or have solutions.
	The three planes meet in a common line: a sheaf	**E1**	States the planes share a single common line, or form a sheaf. No check that the normals are non-parallel is needed.
		[3]	
7(c)(i)	$a > 1$	**B1**	
		[1]	

Question 7 continued

Question	Scheme	Marks	Notes
7(c)(ii)	$\lvert 7(1-a)\rvert(a^2+3) < a^2+3 \Rightarrow \lvert 7(1-a)\rvert < 1$	**M1**	Uses the modulus of their determinant from (a) as the volume scale factor, reaching $\lvert\det \mathbf{A}\rvert < 1$ or both cases. Not for $7(1-a) < 1$ alone.
	$\frac{6}{7} < a < \frac{8}{7}$	**A1**	oe, such as $\left(\frac{6}{7}, \frac{8}{7}\right)$.
		[2]	
	Total for Question 7	**14**	